

Value added problem 1

Name-Archit Shelke

Prn-25070521227

Section-D

Problem Statement:

Given a Binary Search Tree and a range [low, high]. Find all the numbers in the BST that lie in the given range.

Note: Element greater than or equal to root go to the right side.

Concept: Binary Search Tree

Code:

```
#include <stdio.h>
#include <stdlib.h>
typedef struct Node {
    int val;
    struct Node *left, *right;
} Node;
Node* newNode(int val) {
    Node* n = (Node*)malloc(sizeof(Node));
    n->val = val;
    n->left = n->right = NULL;
    return n;
}
Node* insert(Node* root, int val) {
    if (!root) return newNode(val);
    if (val < root->val) root->left = insert(root->left, val);
    else root->right = insert(root->right, val);
    return root;
}
void findInRange(Node* root, int low, int high) {
    if (!root) return;
    if (root->val > low) findInRange(root->left, low, high);
    if (root->val >= low && root->val <= high) printf("%d ", root->val);
    if (root->val < high) findInRange(root->right, low, high);
}
int main() {
    int n, val, low, high;
    Node* root = NULL;
    printf("Name - Archit Shelke\n");
    printf("PRN - 25070521227\n");
    printf("Section - D\n");
    printf("Batch - 2025-29\n\n");
    printf("Enter number of nodes: ");
```

```
scanf("%d", &n);
printf("Enter values: ");
for (int i = 0; i < n; i++) {
    scanf("%d", &val);
    root = insert(root, val);
}
printf("Enter low high: ");
scanf("%d %d", &low, &high);
printf("Output: ");
findInRange(root, low, high);
printf("\n");
return 0;
}
```

Output:

Output	Clear
Name - Archit Shelke PRN - 25070521227 Section - D Batch - 2025-29 Enter number of nodes: 7 Enter values: 24 48 52 17 18 9 7 Enter low high: 8 53 Output: 9 17 18 24 48 52 === Code Execution Successful ===	